

CLASS 04 / SEP 15

MODULE 1 — FIELDS TILING + ARRAYS

BRING YOUR PRECEDENT — WE MODEL IT TODAY

0.1 DUE

1.1 ISSUED TODAY / TILING + ARRAYS

WHERE WE JUST MOVED TO

Module 0 is finished. The ground has changed.

MODULE 0 — DONE

PROCEDURAL

A fixed sequence of operations.

You defined the steps.

One expected result.

MODULES 1–3 — NOW

PARAMETRIC

Relationships between inputs and form.

You define the rules of variation.

A family — you know the range,
not each member.

An array on its own is still procedural — repeat this N times. It becomes parametric the moment a parameter governs how the copies differ.

AN ARRAY IS A LIST OF POSITIONS

Not a command that duplicates things. A list you compute, then place geometry at.

INDEX → POSITION

$i = 0 \rightarrow p = \text{start}$

$i = 1 \rightarrow p = \text{start} + \text{step}$

$i = 2 \rightarrow p = \text{start} + 2 \times \text{step}$

$i = 3 \rightarrow p = \text{start} + 3 \times \text{step}$

$p[i] = \text{start} + i \times \text{step}$

YOU ALREADY KNOW THIS

Move is addition. An array is move, repeated, with the amount driven by the index.

The index is the only new idea: a counter that tells each copy how far along the sequence it is — and therefore how different it should be.

The index is what makes variation possible. Without it every copy is the same copy.

A TILING IS ONE TILE PLUS A RULE

Escher fills a plane with the same shape, over and over. So does a brick wall.

THE TILE

One part, repeated. A panel, a brick, a shingle. Simple on purpose — it is about to be copied a thousand times.

THE RULE

How copies are placed relative to each other. Spacing, offset, rotation, course. Say it in one sentence.

THE SEAM

Where two copies meet. A tile that tiles resolves at the seam. One that merely repeats leaves a gap or a collision.

ALWAYS THE SAME

Every copy is identical. Panels that change shape across a curved surface are Module 3 — a different, harder problem.

IF YOU CANNOT DRAW ONE TILE AND SAY HOW THE NEXT ONE SITS AGAINST IT, YOU DO NOT HAVE A TILING.

FROM FAÇADE TO TILE

You brought the building. Four steps to get from it to one small object.

01

FLATTEN THE ELEVATION

One photograph, straight on. Perspective hides the repeat — correct it or find a drawing.

02

OUTLINE THE REPEATED PART

Slide a rectangle across the façade until it lands on itself. Inside it is the part you are after.

03

DIMENSION IT

Width, height, depth, and how the next one sits against it. Real millimetres, from a drawing or a scale figure.

04

DESIGN YOUR OWN

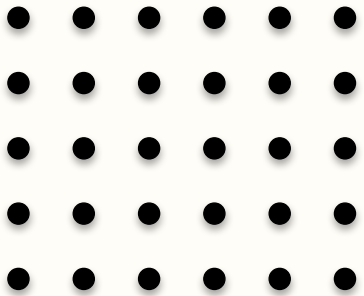
Not a trace. Your tile, driven by theirs. Three to five primitives, at the origin, manifold, closed.

EVERY FACE YOU ADD IS PAID FOR A THOUSAND TIMES. KEEP THE TILE CHEAP.

FOUR ARRAYS, PICK THREE

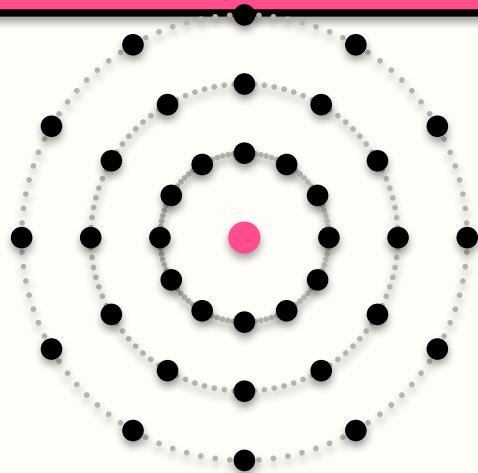
Same tile, four ways of generating the list of positions. You build three.

CARTESIAN



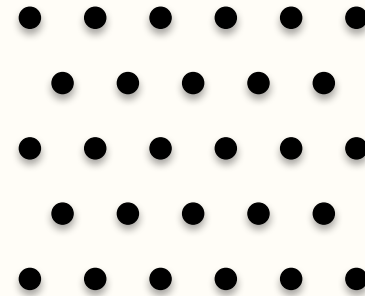
$$p[i][j] = p_0 + (i \cdot dx, j \cdot dy)$$

RADIAL



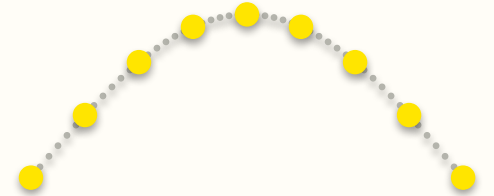
$$p[i] = c + r \cdot (\cos \theta, \sin \theta)$$

HEXAGONAL



offset odd rows by $dx/2$

ON CURVE



$$p[i] = \text{curve at } t = i/n$$

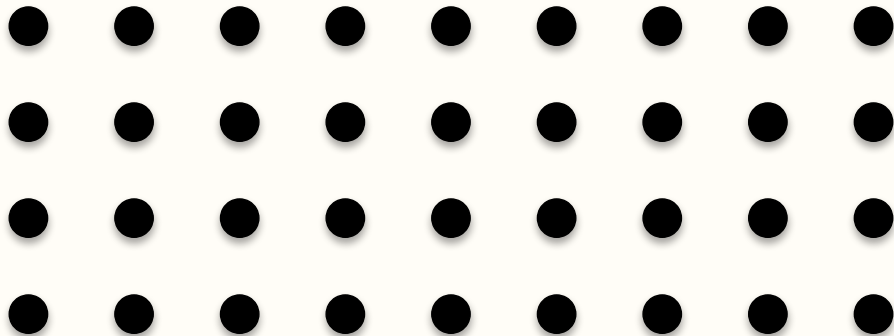
A cartesian array is a list of lists — two counters instead of one. Hold that thought; it returns in Module 3 as a panel field.

THE TILE IS CONSTANT. THE FIELD IS NOT.

The same tile, the same array, twice. One is wallpaper. The other is a system.

PROCEDURAL

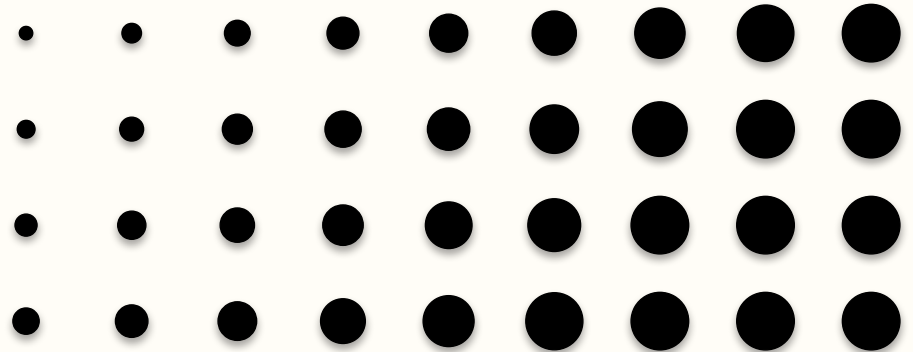
REPETITION



Same tile, same spacing. The index is generated and then ignored.

PARAMETRIC

VARIATION



Same tile — scale, rotation and culling driven by the index.

YOU MAY NOT MODEL A SECOND TILE. EVERYTHING THAT VARIES, VARIES IN THE ARRAY.

TILED FAÇADES

Look for the part and the rule, not the image.

Add: a perforated screen built from one repeated block.

SCREEN / MASHRABIYA

Add: a brick or block wall where the bond is the whole design.

BRICK BOND

Add: a façade of one repeated cast or folded panel.

CAST PANEL

FOR YOUR PRECEDENT, ANSWER THREE QUESTIONS

What is the tile?

What rule places the copies?

Where do two of them meet?

ISSUED TODAY

1.1 — TILING + ARRAYS

DUE CLASS 6 — SEP 29 / 7%

You brought a façade made of one repeated part. Design your own tile from it, model it once, then tile the plane three ways. The precedent drives the tile; you supply the systems that place it and the parameter that makes the field differ. No attractors in this one — that is next week.

STAGE 1

THE PRECEDENT

The building, the repeated part, the rule.
Measured drawing.

STAGE 2

THE TILE

Your design, from theirs. 3-5 primitives, at the origin, manifold.

STAGE 3

THREE ARRAYS

Any three of cartesian, radial, hexagonal, on curve.

ONE TILE. THREE ARRAYS, YOUR CHOICE WHICH. EVERYTHING THAT VARIES, VARIES IN THE ARRAY.

REQUIREMENTS

Counting these earns a C. The rest is judgement.

1	precedent analysis — building, repeated part, rule, in 3–4 sentences with a measured drawing
1	tile designed from that precedent — 3–5 primitives, manifold, at the origin, dimensioned
3	array types built with that tile, from cartesian, radial, hexagonal, along a curve
3	finished compositions — one per array type
1,000+	elements in at least one composition
1+	conditional that removes or alters a subset based on a tested property
1+	array where changing element order visibly changes the result — paired before/after
1+	composition resolved spatially in 3D
2	sheets minimum, 17" × 11", in your visual identity

One tile only. Arrays authored from base components — array add-ons and scatter tools are prohibited.

NOW — DESIGN YOUR TILE

Rest of the session: part dimensioned, tile modeled, first array running.



Precedent approved — a façade made of one repeated part



The part outlined over the elevation, with dimensions



Your tile modeled — one object, at the origin, manifold, 3-5 primitives



Your first array running, count and spacing controllable



One property driven by the index, so the field is not wallpaper

LEAVE WITH A TILE. THE ARRAYS ARE THE EASY PART.

BEFORE NEXT CLASS

WATCH

Tutorial 1.2 — fields and attractors. Distance, falloff, remapping — and reading the field as a 2D image before it drives anything. Sent tonight.

BRING

Your tile and your working arrays. Next week we drive them with attractors, so bring a file you can build on.

READ

Your own 0.1 procedure, out loud, to someone who has not seen the model. Rewrite whatever they cannot follow.

CHOOSE

A site, for next week. You will import it as a point field and diagram it with attractors — pick somewhere you know well enough to have a question about.